Well-Known Continuous Random Variable

1. Uniform Distribution (Continuous)
2. Exponential Distribution
3. Gamma Distribution
4. Beta Distribution
5. Normal Distribution
6. Log Normal Distribution
7. Laplace Distribution

Uniform Distribution

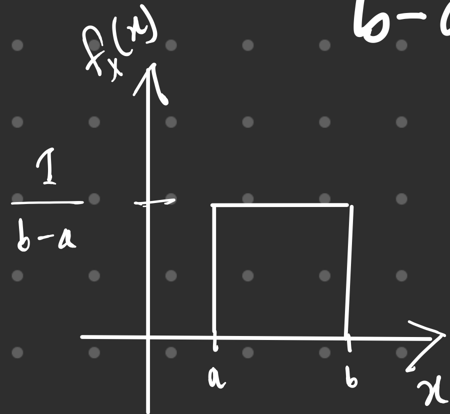
↳ A continuous random variable X is said to follow a uniform distribution on the interval $[a,b]$ if every value in the interval is equally likely to occur. $X \sim \text{Uniform}(a,b)$

Any value from the subinterval is equal

↳ So, probability density remain same.

↳ Probability density depends on the length. → $b-a$

↳ Probability can be more than one depending on length.



∴ total probability = probability density × length

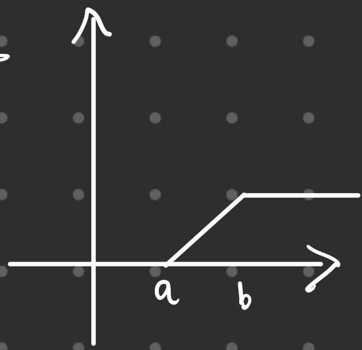
↳ $1 = pd \times b-a \Rightarrow pd = \dfrac{1}{b-a}$

Also, $\displaystyle\int_a^b f(x)\,dx = 1 \Rightarrow cx\Big|_a^b = 1 \Rightarrow c = \frac{1}{b-a}$

$f_X(x) = \begin{cases} c & \text{for } a \le x \le b \\ 0, & o/w \end{cases}$

↳ $f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b \\ 0 & o/w \end{cases}$

Now, $\int_{-\infty}^{x} f_X(x)\,dx = \left[\frac{x}{b-a}\right]_a^x = \frac{x-a}{b-a}$



$\therefore F_X(x) = \begin{cases} 0, & x \le a \\ \frac{x-a}{b-a}, & a < x \le b \\ 1, & x > b \end{cases}$

PDF: $f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & o/w \end{cases}$

CDF: $F_X(x) = \begin{cases} 0, & x \le a \\ \frac{x-a}{b-a}, & a < x \le b \\ 1, & x > b \end{cases}$

$E[x] = \frac{b+a}{2}$

$Var[X] = \frac{(b-a)^2}{12}$

Random Example: Suppose you randomly pick a point along a 20 cm ruler. What's the probability that the point is between 8 cm and 12 cm?

Sol^n: $P(8 \le X \le 12) = \frac{12-8}{20} = \frac{4}{20} = 0.2$

Example 1: Starting at 5:00A.M., every half hour there is a flight from San Francisco airport to Los Angeles International airport. Suppose that none of these planes is completely sold out and that they always have room for passengers. A person who wants to fly to L.A. arrives at the airport at a random time between 8:45A.M.and 9:45A.M.Find the probability that she waits
(a) at most 10 minutes,
(b) at least 15 minutes.

Sol^n: Flights depart every 30 minutes, the waiting time $X \in [0, 30]$ minutes.

So, $X \sim \text{Uniform}(0, 30)$

$\therefore$ PDF : $f(x) = \frac{1}{30}$, for $0 \le x \le 30$.

CDF : $F_X(x) = \frac{x}{30}$; for $0 \le x \le 30$

ⓐ Probability she waits at most 10 minutes —

$$P(X \le 10) = F(10) = \frac{10}{30} = \frac{1}{3}.$$

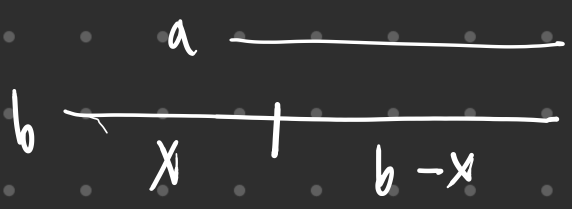
ⓑ Probability she waits at least 15 minutes -

$$P(X \ge 15) = 1 - F(15) = 1 - \frac{15}{30} = \frac{1}{2}.$$

Example 2: lets say there are two sticks of length a and b where b > a, if b is broken down into two pieces randomly, what's the probability that the three of them would make a triangle?

Sol^n: let the three sticks be:

a, X, $b - X$



Applying the triangle inequality to these three:

1. $a + X > b - X \Rightarrow a + 2X > b \Rightarrow X > \frac{b-a}{2}$

2. $a + (b - X) > X \Rightarrow a + b - 2X > 0 \Rightarrow X < \frac{a+b}{2}$

3. $X + (b - X) > a \Rightarrow b > a \rightarrow$ Always true by assumptions

So the values of X that satisfy the triangle inequality are:

$$\frac{b-a}{2} < X < \frac{a+b}{2}$$

$\therefore$ The stick of length b is broken uniformly at random, the probability is uniform on $[0, b]$

$\therefore$ PDF $\rightarrow f_X(x) = \begin{cases} \frac{1}{b}, & 0 < x < b \\ 0, & o/w \end{cases}$

$\therefore$ CDF $\rightarrow \int_{-\infty}^{x} f_X(x)\,dx = \int_{\frac{b-a}{2}}^{x} \frac{1}{b}\,dx = \frac{x - \frac{b-a}{2}}{b}$

$\therefore$ Probability of forming a triangle:

$$F_X\left(\frac{a+b}{2}\right) - F_X\left(\frac{b-a}{2}\right) = \frac{\frac{a+b}{2} - \frac{b-a}{2}}{b} - \frac{\frac{b-a}{2} - \frac{b-a}{2}}{b}$$

$$= \frac{2a}{2b} = \frac{a}{b}$$

Another popular variation of this problem - given a stick of length x, if you randomly two points and break it into three pieces - what's the probability of it being a triangle.

$\hookrightarrow$ Solve it by yourself.

Exponential Distribution

↳ models the time between events ocurring in a process where events happen continuously and inpendepently at a constant average rate.

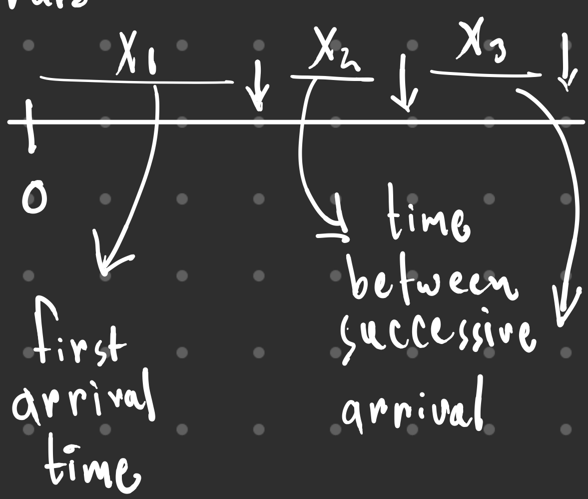
↳ probability of the # of arrivals in a certain period

↳ For such experiments, the number of arrivals is not only random quantity, there is always random quantity of interest.

X_i

— Time between successive intervals

— Interarrival times



$F_X(x) = P[X \leq x]$

So, $\overline{F_X}(x) \rightarrow$ Complementary CDF

$= 1 - F_X(x)$

$= 1 - P[X \leq x] = P[X > x] = P[0 \text{ arrival in } (0, x]] = \dfrac{e^{-\lambda x}(\lambda x)^n}{n!}$

Imagine this scenario,

Customers arrive in a shopping mall following the Poisson distribution

→ at a rate λ customers per hour

→ n customers by time t.

$\dfrac{e^{-\lambda x}(\lambda x)^n}{n!}$

⊛ PDF $\rightarrow f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$

⊛ CDF $\rightarrow F_X(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$

⊛ $E[X] = \frac{1}{\lambda}$, $Var[X] = \frac{1}{\lambda^2}$

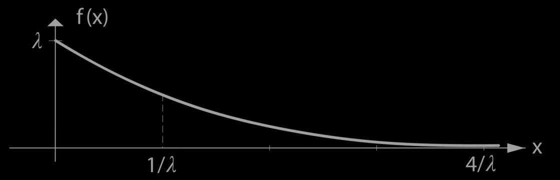


Figure 7.10 Exponential density function with parameter λ.

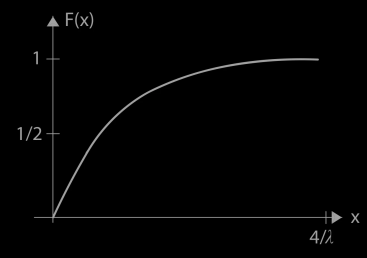


Figure 7.11 Exponential distribution function.

Example 3: Suppose that every three months , on average, an earthquake occurs in California. What is the probability that the next earthquake occurs after three but before seven months?

Soln: Here, $\lambda = \frac{1}{3}$

$$P(3 < X < 7) = F_X(7) - F_X(3)$$
$$= 1 - e^{-\frac{1}{3}\cdot 7} - 1 + e^{-\frac{1}{3}\cdot 3}$$
$$= 0.271$$

Example 4: Suppose the average lifetime of a lightbulb is 1,000 hours. Assume the lifetime follows an exponential distribution. What is the probability that a bulb lasts more than 1,200 hours?

Soln:

$$\lambda = \frac{1}{1000} = 0.001 \text{ per hour}$$

$$P(X > 1200) = 1 - F(1200) = e^{-0.001 \times 1200} = e^{-1.2}$$
$$\approx 0.301$$

Memoryless property of exponential distribution:

Gamma Distribution

Before we dive into gamma distribution, let us go through a few things

What does exponential distribution do? → $f_X(x) = \lambda e^{-\lambda x}$

↳ finds the time to get the first arrival. $F_X(x) = 1 - e^{-\lambda x}$

↳ corresponds to Geometric Distribution

As for Gamma distribution (aka n-Erlang Distribution)

↳ finds the time to get the k^{th} arrival

↳ corresponds to Pascal Distribution

→ IID

identically and independently distributed

$$X_1 + X_2 + X_3 + \dots + X_n = X$$

↳ Formal definition: Models the waiting time until the occurrence of multiple events in a Poisson process.

↳ Gamma Distribution has two parameters.

1. Shape Parameter (n) → number of events you're waiting for.
2. Scale Parameter (θ) → average gap between successive arrivals

Here, $\theta = \frac{1}{\lambda}$; λ = rate parameter.

↳ if $n=1$, $Gamma(1,\lambda) \equiv Exponential(\lambda)$

∴ PDF : $f_X(x) = \begin{cases} \lambda e^{-\lambda x} \frac{(\lambda x)^{n-1}}{(n-1)!} & \text{if } x \geq 0 \\ 0 & \text{o/w} \end{cases}$

CDF : $F_X(x) = P(X \leq x) = \sum_{i=n}^{\infty} \frac{e^{-\lambda x} (\lambda x)^i}{i!}$

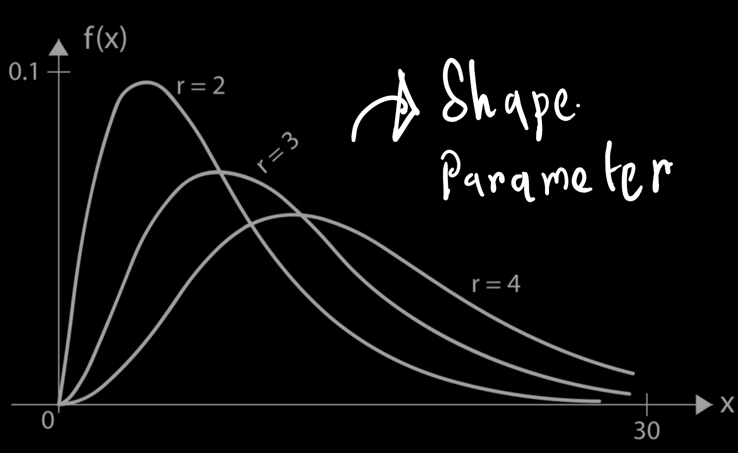


Figure 7.12 Gamma densities for $\lambda = 1/4$.

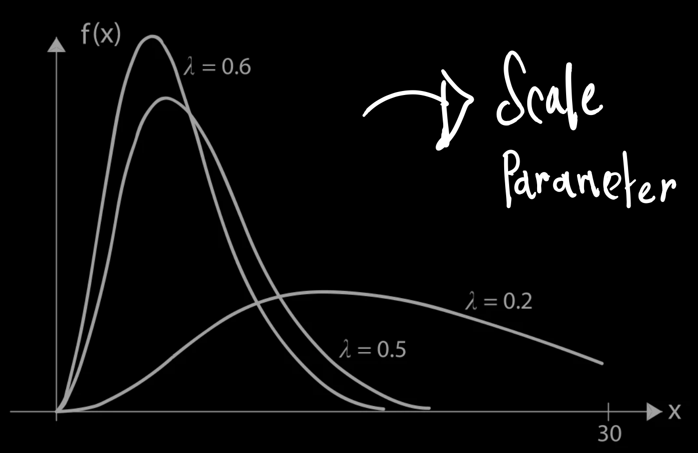


Figure 7.13 Gamma densities for $r = 4$.

For gamma distribution → $E(X) = \frac{r}{\lambda}$, $Var(X) = \frac{r}{\lambda^2}$, $\sigma_x = \frac{\sqrt{r}}{\lambda}$

Example 5: Customers arrive at a restaurant at a Poisson rate of 12 per hour. If the restaurant makes a profit only after 30 customers have arrived, what is the expected length of time until the restaurant starts to make profit?

Solⁿ: Here, $\lambda = 12$ per hour

after 30 customers, $n = 31$

$$\therefore E[X] = \frac{n}{\lambda} = \frac{31}{12} = 2.583 \text{ hours}$$
$$= 2 \text{ hours } 35 \text{ mins}$$

Gamma Function

↳ if $n > 0$, then the integral

$$\Gamma n = \int_0^\infty x^{n-1} e^{-x} \, dx = (n-1)!$$

Also, $\Gamma(n+1) = n!$

Then,

$$f_X(x) = \begin{cases} \dfrac{\lambda e^{-\lambda x} (\lambda x)^{r-1}}{\Gamma(r)} & \text{if } x \geq 0 \\ 0 & \text{o/w} \end{cases}$$

Beta Distribution

Watch this: → https://youtu.be/juF3r12nM5A?si=XZY73wy6MXVlYAnO

Beta Function:

$$B(\alpha,\beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\,dx$$

$$\Rightarrow B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

$$B(\alpha,\beta) = B(\beta,\alpha)$$

$$B(\alpha,\beta) = \frac{(\alpha-1)!\,(\beta-1)!}{(\alpha+\beta-1)!}$$

Intuitive Understanding

Think of the Beta function as measuring how much probability mass is between 0 and 1, shaped by two forces:
α pushing the curve right, and
β pushing it left.

In the Beta distribution, we're modeling a random variable that lies between 0 and 1, such as:

-> The probability of success
-> The proportion of time something happens
-> The chance a coin lands heads

🔍 What is the Beta Distribution, Intuitively?

The Beta distribution is a way to describe a **probability about a probability.**

That might sound weird — but it's very useful.
Let's say you **don't know** the probability of some event, but you want to model your **belief** about what it might be. That's where Beta comes in.

🧠 **Think of it like this:**

> "I think the probability of success (e.g. a user clicking a button) is probably around 70%, but I'm not too sure."

You want to represent that "not too sure" part too — the **uncertainty** about the 70%. Beta helps do this.

🎯 What Does It Give Us?

If $p \sim \text{Beta}(\alpha, \beta)$, it means:

- The **mean** (average guess for the probability) is

$$\mu = \frac{\alpha}{\alpha+\beta}$$

- The **more data** you have (higher α and β), the more **confident** (less spread) you are in your estimate.

📦 Example 1: Coin Flip

Imagine you have a **biased coin**, but you don't know the bias.

- You flip it 10 times: 7 heads, 3 tails.
- You want to estimate the **true probability of heads**.

You model this with:

$$\text{Beta}(7+1, 3+1) = \text{Beta}(8, 4)$$

This gives you:

- An **average estimate** of $\frac{8}{8+4} = \frac{2}{3}$
- A **distribution** showing how confident you are (spread around 0.66)

This is better than just saying "the probability is 0.7" — it tells how **uncertain** you are too.

🎮 Example 2: A/B Testing

You show **Version A** of your website to 100 users. 20 of them click.
You show **Version B** to 100 users. 25 click.

Instead of just saying:

- A: 20% CTR
- B: 25% CTR

You use:

- A: $\text{Beta}(21, 81)$
- B: $\text{Beta}(26, 76)$

You now get a **probability distribution** over each version's conversion rate.
You can even calculate:

> What's the chance that B is better than A?

Answer: Beta gives you a **full probabilistic comparison**, not just point estimates.

↳ PDF: $f(x;\alpha,\beta) = \begin{cases} \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}; & 0<x<1 \\ 0 & ; \text{o/w} \end{cases}$

↳ CDF: $F(x;\alpha,\beta) = \int_0^x f(t;\alpha,\beta)\,dt = I_x(\alpha,\beta)$

Regularized Incomplete Beta Function

↳ $E[X] = \frac{\alpha}{\alpha+\beta}$; $Var[X] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

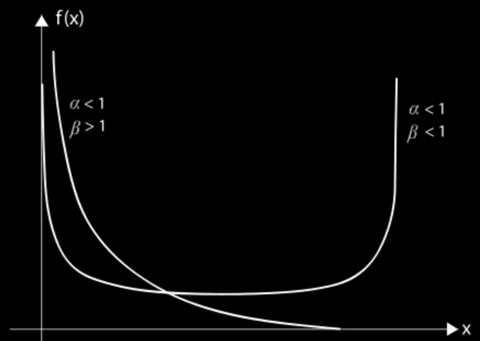


Figure 7.14 Beta densities for $\alpha < 1, \beta < 1$ and $\alpha < 1, \beta > 1$.

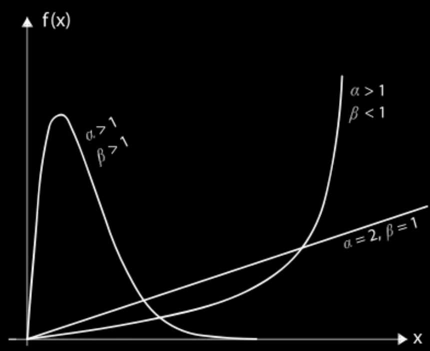


Figure 7.15 Beta densities for the indicated values of α and β.

Example C: The proportion of the stocks that will increase in value tomorrow is a beta random variable with parameters α and β. Today, these parameters were determined to be α = 5 and β = 4, by a financial analyst, after she assessed the current political, social, economical, and financial factors. What is the probability that tomorrow the values of at least 70% of the stocks will move up?

Solⁿ:

Given $\alpha = 5$, $\beta = 4$.

$$P(X \geq 0.7) = 1 - P(X < 0.7)$$

$$= 1 - \int_0^{0.7} \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}\,dx$$

$$= 1 - 280 \times 2.8782 \times 10^{-3} = 0.194$$

Example 7: A beam of length l, rigidly supported at both ends, is hit suddenly at a random point. This leads to a break in the beam at a position X units from the right end. If X/l is beta with parameters α = β = 3, find E(X), Var(X), and P(l/5 < X < l/4).

Solnᵒ:

Here, $X/l \sim \text{Beta}(3,3)$. So, $Y = X/l$. So, $X = lY$.

Now, $E[X] = E[lY] = l\,E[Y] = l\dfrac{\alpha}{\alpha+\beta} = \dfrac{l}{2}$.

$$Var[X] = l^2[Var\,Y] = l^2\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{3.3}{6^2.7}l^2 = \frac{l^2}{28}$$

$$P(l/5 < X < l/4) = P(1/5 < Y < 1/4)$$

$$= F(1/4) - F(1/5)$$

$$= \frac{1}{B(3,3)} \times \int_{1/5}^{1/4} x^2(1-x)^2\,dx$$

$$= 0.0456$$

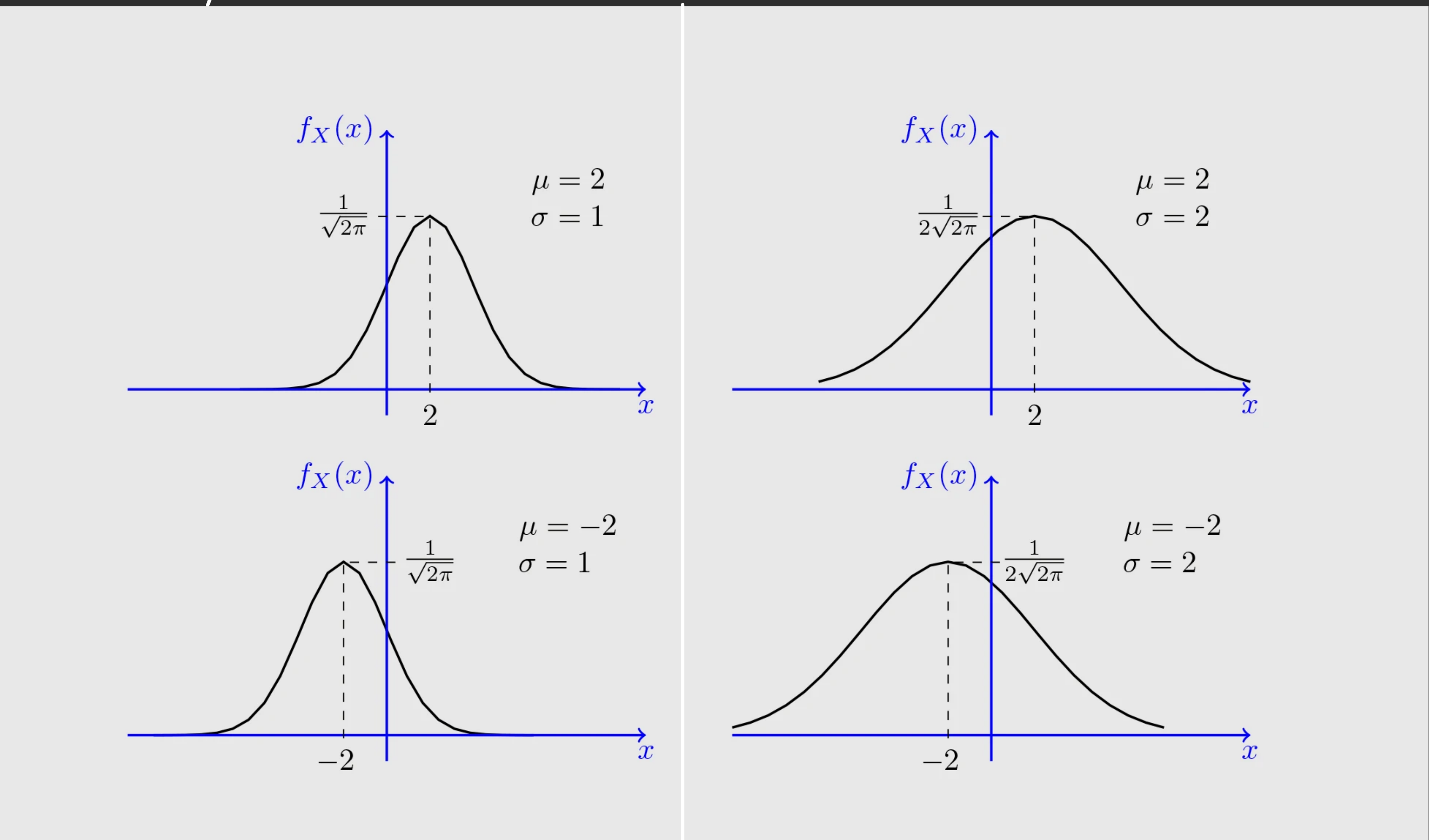
Gaussian / Normal Distribution

↳ Watch this → https://youtu.be/xlxaa9YhT6A?si=yRowgCzqrUzbGdOT

↳ Characterized by its bell shaped curve

↳ Two parameters → Mean (μ) → Determines the center of the distribution
→ Standard Deviation (σ) → Determines the spread

↳ $N \sim (\mu, \sigma^2)$



↳ Some example of Bell Shaped Curve

↳ PDF : $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$

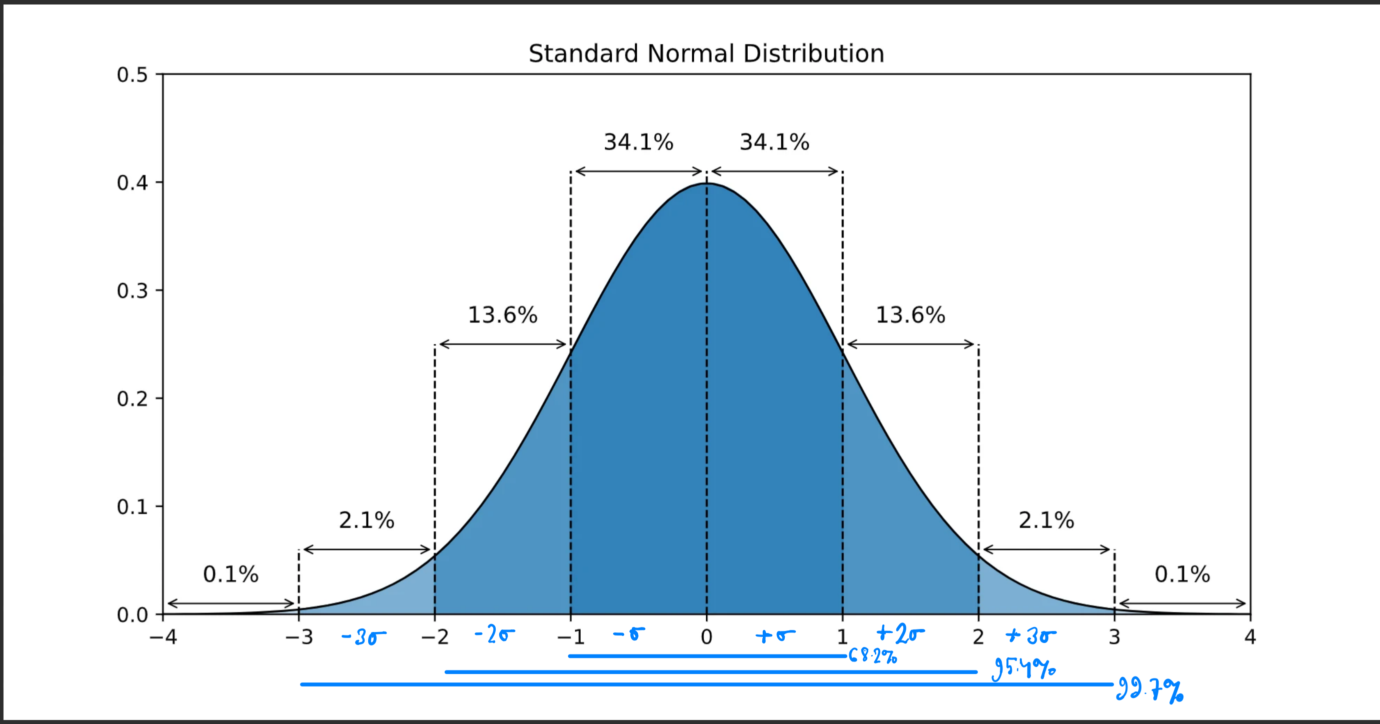
↳ CDF : $F_X(x) = \int_{-\infty}^{x} f(x)\,dx$

↳ $E[X] = \mu, \quad Var[X] = \sigma^2$

Standardization of Gaussian/Normal Distribution

↳ For this case, $\mu = 0, \sigma^2 = 1$

↳ To convert a normal variable to a standard normal variable, $Z = \frac{X-\mu}{\sigma} \Rightarrow Z \sim N(0,1)$

↳ 

↳ ⊛ PDF: $f(z) = \varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$

↳ Gives the height of the bell curve at a point z

⊛ CDF: $F(z) = \Phi(z) = P(Z \leq z) = \int_{-\infty}^{z} \varphi(z)\,dz$

↳ Gives the area under the curve to the left of z

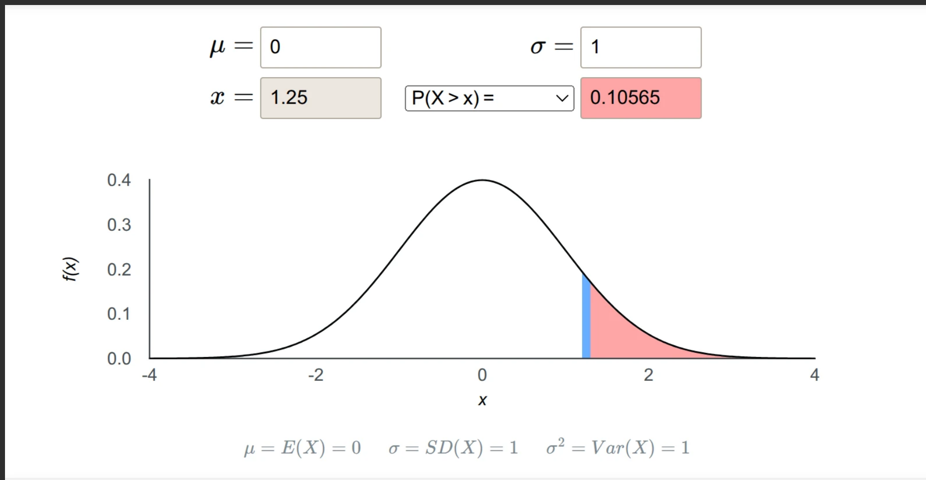
⊛ Complementary CDF: $\bar{F}(z) = 1 - F(z) = P(Z > z) = 1 - \Phi(z)$

⊛ $E[z] = 0$, $Var[z] = 1$

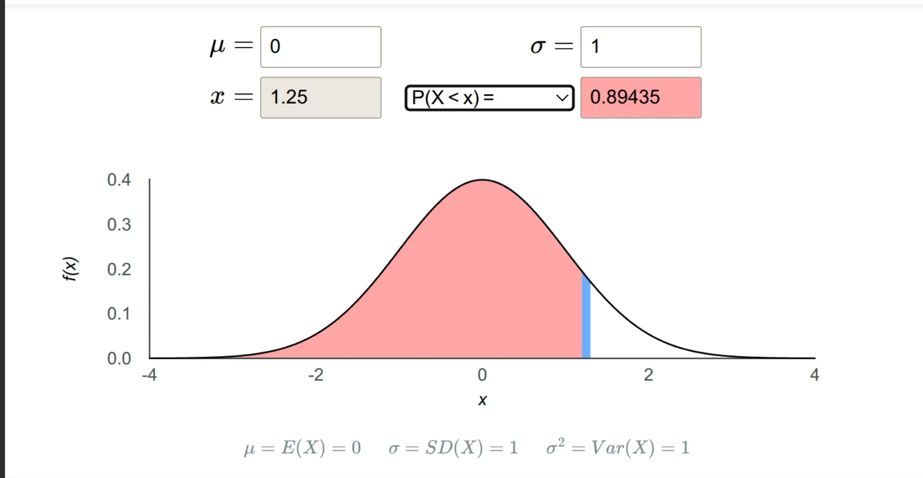
Let's take a look at a few more diagrams:

Assuming, $x = 1.25$

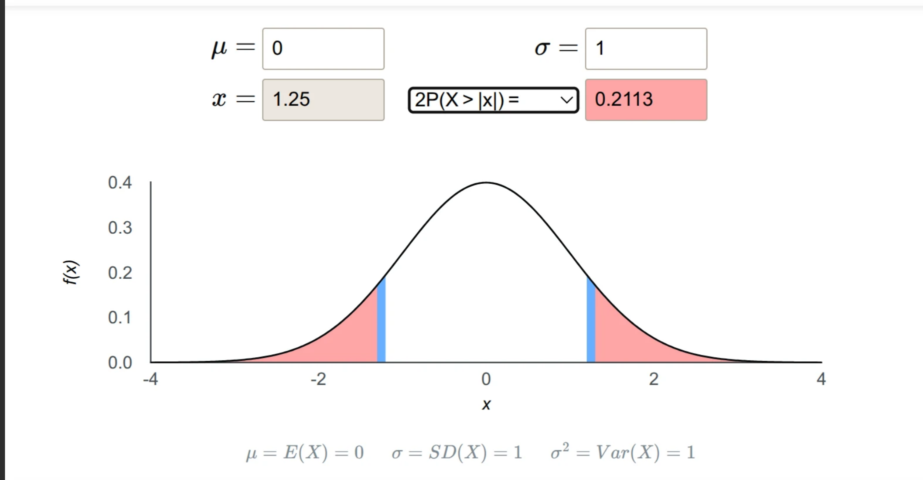
$$P(X > x) \leftarrow$$
$$= 1 - P(X \leq x)$$
$$= 1 - \phi(x)$$



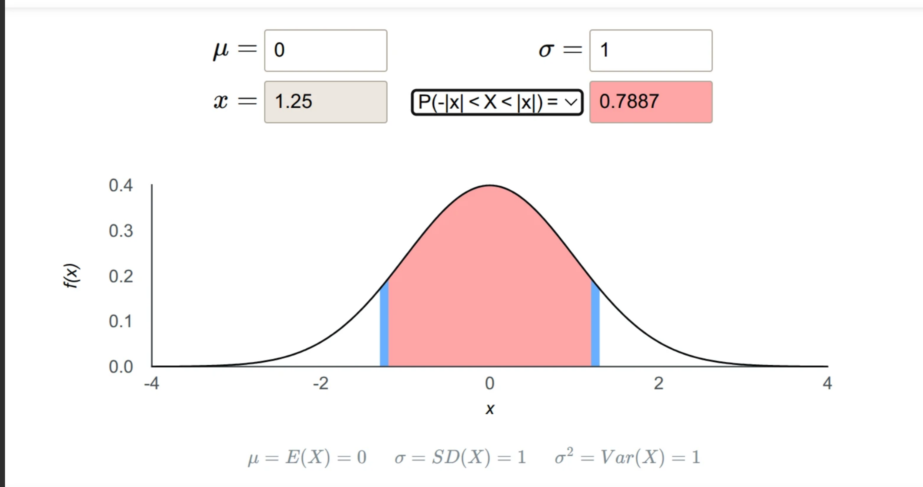
$$P(X < x) \leftarrow$$
$$= P(X \leq x)$$
$$= \phi(x)$$



$$2P(X > |x|) \leftarrow$$
$$= 2(1 - \phi(x))$$



$$P(-x < X < x) \leftarrow$$
$$= \phi(x) - \phi(-x)$$
$$= \phi(x) - 1 + \phi(x)$$
$$= 2\phi(x) - 1$$



Z - Table Values

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997

From the table:

$\phi(1) = 0.84134$

$\phi(0) = 0.50000$

$\phi(-z) = 1 - \phi(z)$

$\phi^{-1}(-p) = -\phi^{-1}(1-p)$

$p = \phi(z)$

$\phi^{-1}(p) = z$

Example: Find the following probabilities:

1. $P[Z > 1.26]$
2. $P[Z < -1.26]$
3. $P[Z > -1.26]$
4. $P[-1.26 < Z < 1.26]$
5. $P[Z \leq 4.6]$
6. Find the value of z such that $P[Z > z] = 0.05$
7. Find the value of z such that $P[-z < Z < z] = 0.99$

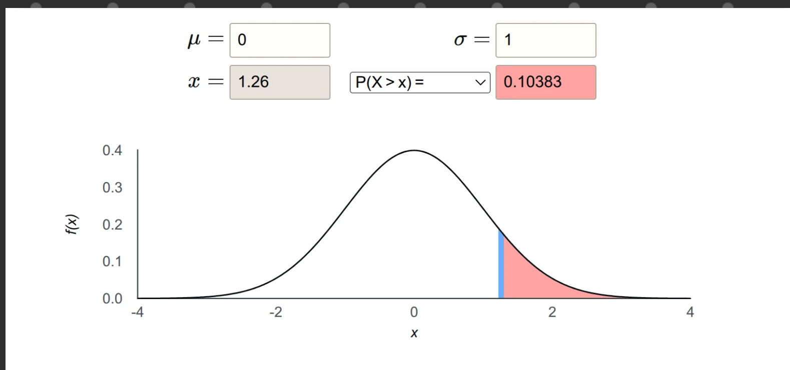
Sol^n:

① $P[Z > 1.26]$

$= 1 - P[Z \leq 1.26]$

$= 1 - 0.8926$

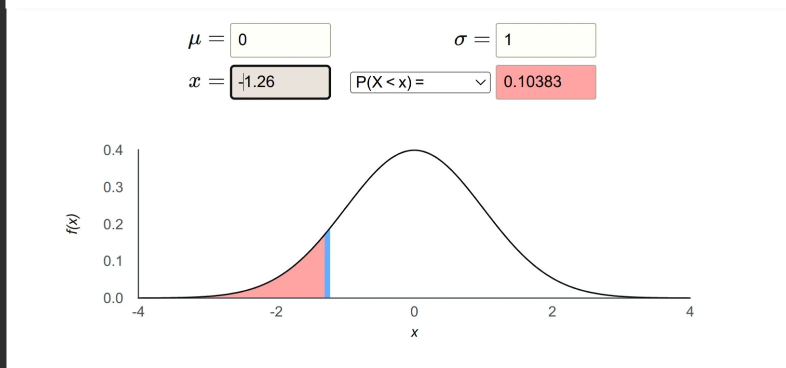
$= 0.1074$



② $P[Z < -1.26]$

$= 1 - P[Z < 1.26]$

$= 0.10383$

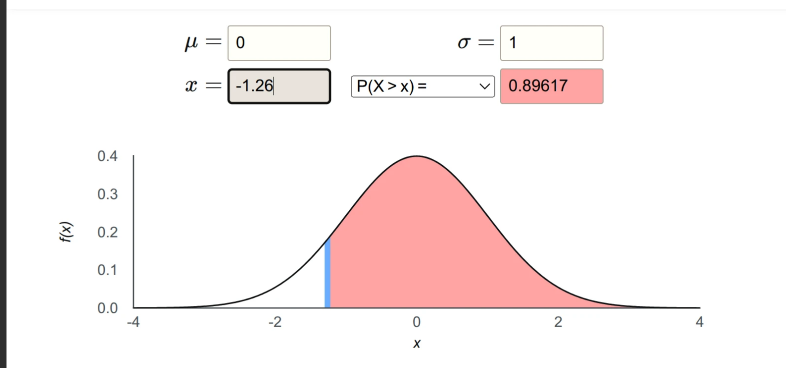


③ $P[Z > -1.26]$

$= 1 - P[Z > 1.26]$

$= 1 - 1 + P[Z \leq 1.26]$

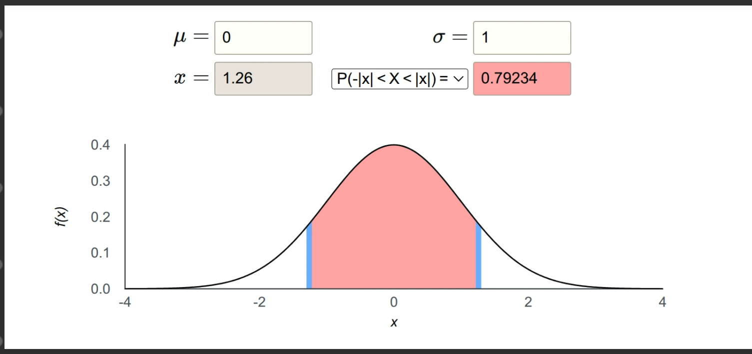
$= 0.89617$



4. $P[-1.26 < Z < 1.26]$

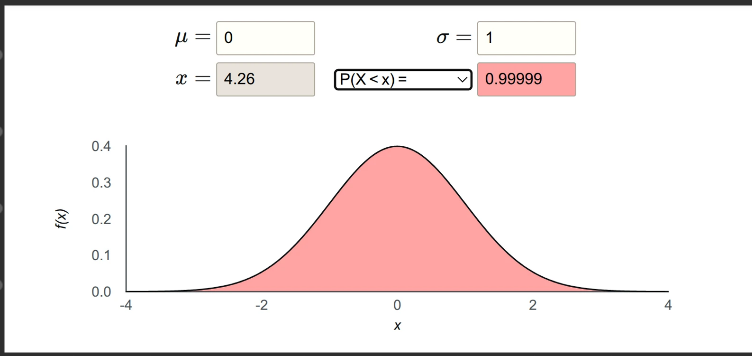
$= 2\Phi(1.26) - 1$

$= 0.79234$



5. $P[Z \leq 4.6]$

≈ 1

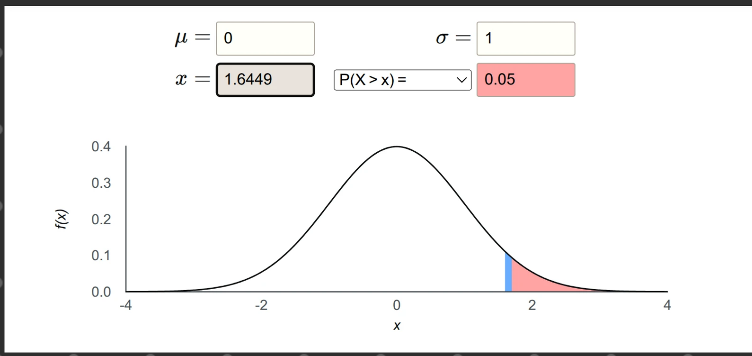


6. $P[Z > z] = 0.05$

$\Rightarrow P[Z \leq z] = 0.95$

$\hookrightarrow \Phi^{-1}(p) = z$

$\Rightarrow \Phi^{-1}(0.95) = 1.645$



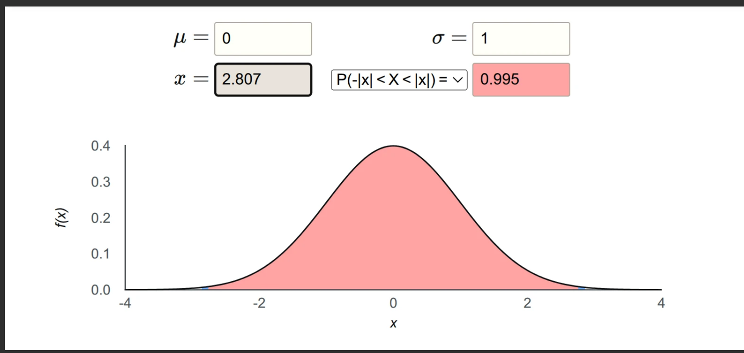
7. $P[-z < Z < z] = 0.99$

$\hookrightarrow 2\Phi(z) - 1 = 0.99$

$\hookrightarrow \Phi(z) = 0.995$

$\hookrightarrow z = \Phi^{-1}(0.995)$

$= 2.87$



* Answer may vary depending on the z-table you are given.

Example: Suppose that the current measurement in a strip of wire are assumes to follow a normal distribution with a mean of 10 milliamperes and a variance of 4. What is the probability that a measurement exceeds 13 milliamperes?

Sol^n: Let X denote the current in milliamperes.

$P[X>13] = ?$ $\quad Z = \frac{X-\mu}{\sigma} = \frac{13-10}{2} = 1.5$

$\therefore P[X>13] = P[Z>1.5] = 0.9332.$

Example: Heights of Men and Women. Suppose that heights in inches, of the women in a certain population follow the normal distribution with mean 65 and standard deviation 1. And that the heights of the men follow the normal distribution with mean 68 and standard deviation 3. Suppose also that one woman is selected at random and independently, one man is selected at random. We shall determine the probability that the woman will be taller than the man.

Sol^n: Given, $W \sim N(\mu_W = 65, \sigma_W^2 = 1^2 = 1)$

$M \sim N(\mu_M = 68, \sigma_M^2 = 3^2 = 9)$

Let, $X = W - M \Rightarrow X \sim N(\mu_X, \sigma_X^2)$

$\mu_X = \mu_W - \mu_M = 65 - 68 = -3$

$\sigma_X^2 = \sigma_W^2 + \sigma_M^2 = 1 + 9 = 10$

So, $X \sim (-3, 10)$

Standardizing X

$Z = \frac{X-\mu_X}{\sigma_X} = \frac{0-(-3)}{\sqrt{10}} = \frac{3}{\sqrt{10}} \approx 0.94$

Now, $P(Z>0.9487) = 1 - P(Z \leq 0.9487)$

$= 1 - \Phi(0.9487)$

$= 1 - 0.8289$

$= 0.1711$

$\therefore P(\text{woman taller than man}) = 17.11\%$

Example:

Heights of Men and Women. Suppose that heights in inches, of the women in a certain population follow the normal distribution with mean 65 and standard deviation 1. And that the heights of the men follow the normal distribution with mean 68 and standard deviation 3. Suppose also that one woman is selected at random and independently, one man is selected at random. We shall determine the probability that the woman will be taller than the man.